AUTOMOBILE SELLING PRICE PREDICTION USING RANDOM FOREST REGRESSION AND FEATURE ENGINEERING

[U21ADP05] Exploratory Data Analysis and Visualization

REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Estimating the fair selling price of a used automobile is difficult because the value depends jointly on age, mileage, ownership history, manufacturer, model and engine characteristics. This project develops a supervised machine learning system that predicts the selling price of used automobiles from tabular vehicle records, treating price estimation as a regression problem.

The dataset contains 5,500 records with 18 attributes, nine of which have between 782 and 855 missing values. The workflow applies median and mode imputation, exploratory visualisation, three engineered features (Car_Age, Mileage_per_Engine and First_Owner), regression-based feature selection using correlation, F-score and mutual information, and a scikit-learn pipeline with one-hot encoding. The derived feature Price_per_HP was deliberately excluded to prevent target leakage.

Linear Regression, Decision Tree Regression (maximum depth 10) and Random Forest Regression (200 trees) were trained on an 80:20 split of 4,400 training and 1,100 test records and compared using MAE, RMSE and R^2. Random Forest performed best, with an MAE of 1349.7979, an RMSE of 2113.4888 and an R^2 of 0.9716, reducing MAE by about 60% relative to Linear Regression and 32% relative to the Decision Tree. Feature-importance analysis showed that vehicle age and manufacture year dominate the predictions, followed by premium manufacturers and mileage. Actual-versus-predicted and residual plots show close agreement overall, with larger errors for high-priced vehicles.

The study is limited by a single train–test split and by imputation performed before splitting; cross-validation, leakage-free preprocessing and external validation are suggested as future work. The results indicate that ensemble tree models are a reliable basis for automobile price estimation on this dataset.

Keywords: *automobile price prediction, machine learning, Random Forest regression, feature engineering, regression analysis, used vehicles.*

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
CV	Computer Vision
ML	Machine Learning
RF	Random Forest
DT	Decision Tree
LR	Linear Regression
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination
EDA	Exploratory Data Analysis
MI	Mutual Information
CSV	Comma-Separated Values
HP	Horsepower
GPU	Graphics Processing Unit
SUV	Sport Utility Vehicle
AWD / FWD	All-Wheel Drive / Front-Wheel Drive

CHAPTER 1
INTRODUCTION

1.1 Objective

The aim of this project is to design, implement and evaluate a machine learning system that predicts the selling price of a used automobile from its recorded attributes (manufacturer, model, year, mileage, ownership history and engine characteristics). The task solved is **supervised regression on structured (tabular) data**, where the target, Selling_Price, is a continuous number. The specific objectives are:

- to inspect and clean a dataset of 5,500 records with missing values in nine attributes;
- to derive features (Car_Age, Mileage_per_Engine, First_Owner) while avoiding target leakage and to rank features using correlation, F-score and mutual information;
- to train Linear, Decision Tree and Random Forest regression models on the same split; and to compare them using MAE, RMSE and R^2 and interpret the best model through residuals and feature importance.

1.2 Overview

The used-automobile market contains vehicles of widely different age, brand, condition and specification, so a single rule rarely gives a fair price, and a wrong price makes a seller lose money or a buyer overpay. Machine learning offers a systematic alternative: a model learns the relationship between vehicle attributes and past selling prices and then estimates the price of a new vehicle. Because price is continuous, regression algorithms are the natural choice. Ensemble methods such as Random Forest [1] combine many decision trees and work well on mixed numerical and categorical data, and comparative studies of resale-price forecasting have found Random Forest particularly effective [2].

In this project the dataset is loaded in Google Colab, inspected and cleaned by median and mode imputation, and explored with plots. A scikit-learn pipeline performs imputation and one-hot encoding, after which Linear, Decision Tree and Random Forest regressors are trained on an 80:20 split. The models are compared using MAE, RMSE and R^2, and the best one is examined through actual-versus-predicted plots, residuals and feature importance.

1.3 Scope

The project covers the complete workflow for one tabular dataset, from loading to interpretation. Although this report follows the Computer Vision project report format, the data are structured records rather than images; the same discipline of dataset description, preprocessing, model training, metrics and error analysis is applied. The boundaries of the work are:

- **Dataset:** one dataset of 5,500 records and 18 attributes; no external dataset.
- **Environment:** Python on Google Colab with pandas, scikit-learn, matplotlib, seaborn and Plotly.
- **Users:** buyers, sellers and dealers who need a data-driven starting price.
- **Limitations:** no hyperparameter tuning, cross-validation, image-based or time-series modelling, or web deployment; findings hold for this dataset and configuration.

CHAPTER 2
LITERATURE SURVEY

2.1 Review of Related Work

Research on used-vehicle valuation spans statistical regression, supervised learning, feature engineering and ensemble methods. Table 2.1 summarises twelve relevant studies in chronological order. Where the dataset is marked "see source", only the published title and approach are summarised; the dataset details and metrics are not reproduced here.

Table 2.1 Summary of the reviewed literature

Authors, year	Technique	Dataset	Reported result	Limitation
Lessmann and Voß, 2017 [2]	Comparison of many regression methods	Used-car resale-price data	Random Forest particularly effective; linear regression discouraged	Tied to the data used
Wang et al., 2021 [3]	Supervised learning algorithms	Used-car data (see source)	Data-driven price estimation is feasible	Dataset-dependent results
Bies et al., 2021 [4]	Random Forest to reduce information asymmetry	Used-car data (see source)	Supports Random Forest for price estimation	Single-model focus
Arora et al., 2022 [5]	Evaluation of ML models for resale value	Used-car data (see source)	Compares predictive proficiency of models	Metrics not comparable across datasets
Bukvić et al., 2022 [6]	Linear regression versus classification	Croatian online listings; two sets, three months apart	Average price rose about EUR 1,391 between the sets	Single market; limited model diversity
Cui et al., 2022 [7]	ResNet with iterative XGBoost + LightGBM	2021 MathorCup competition data	More accurate than Random Forest and ResNet alone	Complex; lower interpretability
Varshitha et al., 2022 [8]	Neural networks and ML models	Used-car data (see source)	Evaluates neural and classical models	Needs more data and tuning
Sharma and Mitra, 2024 [9]	Multivariate Adaptive Regression Splines	Used-car data (see source)	Flexible regression suits price modelling	Depends on spline settings
Kadir et al., 2025 [10]	ML versus deep learning	Used-car data (see source)	Compares classical and deep models	Deep models need more data
Vanjari et al., 2025 [11]	Stacked regression ensemble	Used-car data (see source)	Stacking for pre-owned car prices	Higher complexity
Bergmann and Feuerriegel, 2025 [12]	ML with granular equipment data	92,239 real-world sales	Equipment data improved MAE by 3.27%	Needs detailed equipment records
Fayyaz et al., 2025 [13]	Feature engineering; 10+ models incl. stacking	Kaggle: 4,009 records, 12 features	Stacking R^2 rose from 0.14 to 0.8899	Small single-source dataset

2.2 Research Gap

The reviewed studies show that tree-based ensembles, careful feature engineering and richer input data all improve used-car price prediction, but each reports results on a different dataset, so numbers cannot be compared directly. Several studies rely on complex multi-model frameworks or detailed equipment data that are not available in ordinary listings, and few state explicitly how target leakage is avoided or how the fitted model is interpreted. This project addresses the gap in a practical way: it compares three interpretable regressors on an identical split of a moderately sized, partly incomplete dataset, excludes the leaking feature Price_per_HP, and analyses residuals and feature importance, while openly stating the limitations of the evaluation.

CHAPTER 3
PROBLEM IDENTIFICATION AND METHODOLOGY

3.1 Problem Identification

The selling price of a used automobile depends jointly on age, mileage, number of owners, accident and service history, brand, model, body type and engine characteristics. These factors interact: for example, a luxury brand loses value differently from an economy brand, and the effect of mileage depends on age. Manual valuation is therefore subjective and inconsistent, and an incorrect price harms both parties. In addition, real datasets are imperfect. In the dataset used here, nine of the eighteen attributes have between 782 and 855 missing values each, and several attributes are categorical, so a model cannot be trained on the raw table. Table 3.1 lists the shortcomings of common approaches and how this project addresses them.

Table 3.1 Shortcomings of existing approaches and the treatment adopted

Shortcoming of existing approaches	Treatment in this project
Manual or rule-based valuation is subjective and inconsistent	Data-driven model trained on 5,500 historical records
Linear models cannot capture non-linear depreciation [2]	Random Forest used as the main model; Linear Regression kept only as a baseline
Raw data contain missing values and text categories	Median and mode imputation; one-hot encoding inside a pipeline
Derived features built from the target (e.g., Price_per_HP) leak information	Price_per_HP deliberately excluded from the feature set
Single-model results give no basis for comparison	Three models evaluated on the same 80:20 split with the same metrics
Accuracy alone does not explain the predictions	Residual plot, actual-versus-predicted plot and feature importance

3.2 Methodology

The methodology, shown in Fig 3.1, is a sequential workflow. Each stage feeds the next, and the decision points concern missing values and the choice of the best model.

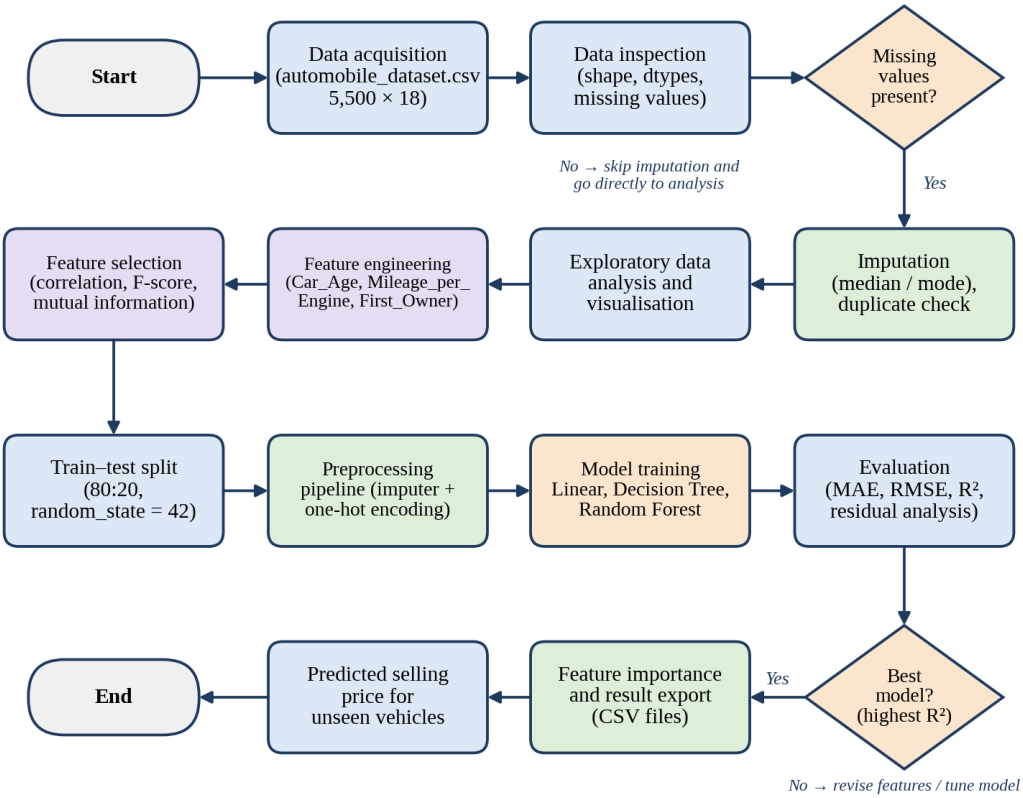


Fig 3.1 Flow chart of the methodology

The eight stages are described briefly below; Chapter 4 gives the details.

1. **Data acquisition and inspection:** load the CSV file, check shape, data types, duplicates and missing values.
2. **Cleaning:** impute numerical gaps with the median and categorical gaps with the mode.
3. **Exploratory analysis:** histograms, bar and pie charts, scatter plots and a correlation heatmap.
4. **Feature engineering:** create Car_Age, Mileage_per_Engine and First_Owner.
5. **Feature selection:** rank features by absolute correlation, F-score and mutual information.
6. **Splitting and preprocessing:** 80:20 split followed by an imputation and one-hot encoding pipeline.
7. **Model training:** Linear Regression, Decision Tree Regression and Random Forest Regression.
8. **Evaluation and output:** MAE, RMSE and R^2, residual analysis, feature importance and CSV export.

CHAPTER 4
PROPOSED SYSTEM

4.1 System Architecture Diagram

The proposed system is organised into five layers, as shown in Fig 4.1. The data layer supplies the raw records, the data preparation layer cleans and enriches them, the modelling layer trains the regressors, the evaluation layer measures and explains their behaviour, and the output layer delivers predictions and saved results.

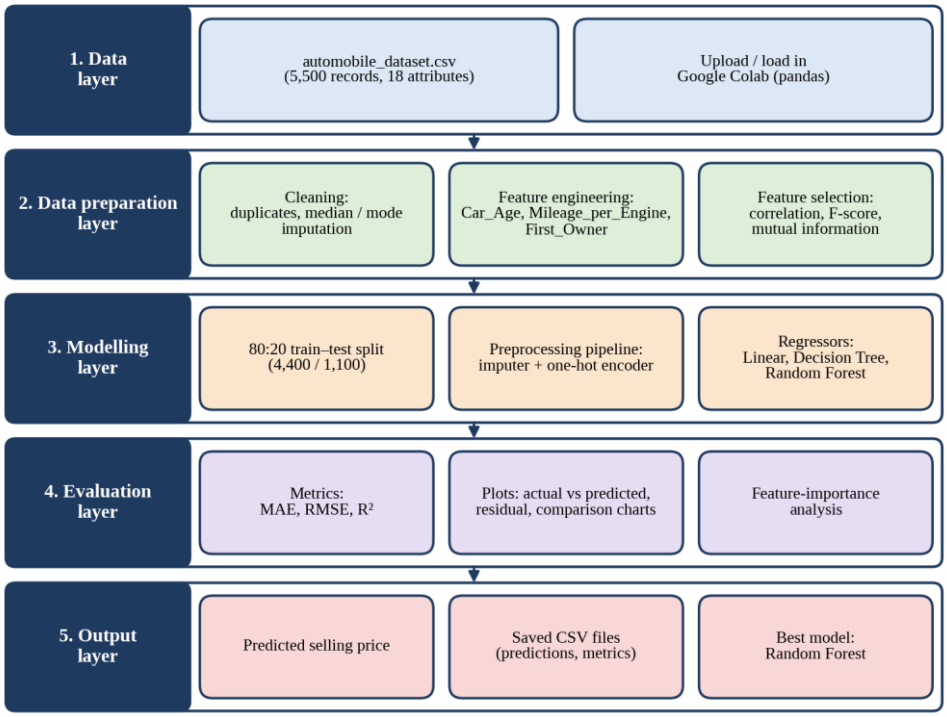


Fig 4.1 System architecture diagram

4.2 Dataset Description

The dataset, automobile_dataset.csv, is loaded into Google Colab and contains 5,500 automobile records described by 18 attributes. It is purely tabular, so image resolution does not apply. The target label is the numerical attribute Selling_Price. Table 4.1 gives the specification, and Table 4.2 describes each attribute together with its data type and the number of missing values found during inspection.

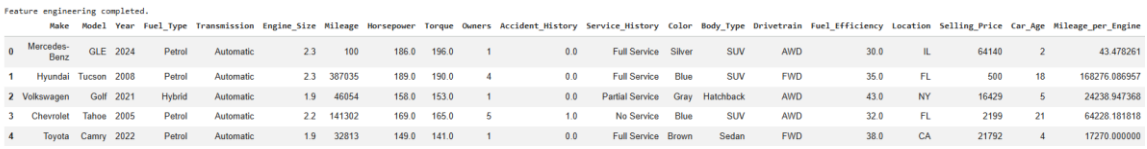
Table 4.1 Dataset specification

Item	Description
Source file	automobile_dataset.csv (uploaded to Google Colab)
Records / attributes	5,500 records; 18 original attributes (20 input features after engineering)
Data type	Structured tabular data (9 numerical and 9 categorical original attributes)
Target variable	Selling_Price (continuous, integer)
Duplicate records	0
Attributes with missing values	9 attributes; 7,316 missing cells (7.4% of 99,000 cells)
Train / test split	80:20, random_state = 42 (4,400 training and 1,100 test records)
Annotation format	Numeric price label per record; no manual annotation required

Table 4.2 Attribute description of the dataset

Attribute	Data type	Description	Missing	Missing %
Make	object	Vehicle manufacturer	0	0.0%
Model	object	Vehicle model name	0	0.0%
Year	int64	Year of manufacture	0	0.0%
Fuel_Type	object	Petrol, Diesel, Hybrid or Electric	0	0.0%
Transmission	object	Automatic or Manual	827	15.0%
Engine_Size	float64	Engine displacement	844	15.3%
Mileage	int64	Distance driven	0	0.0%
Horsepower	float64	Engine power	800	14.5%
Torque	float64	Engine torque	799	14.5%
Owners	int64	Number of owners	0	0.0%
Accident_History	float64	Accident record indicator	812	14.8%
Service_History	object	Full, partial or no service record	855	15.5%
Color	object	Exterior colour	803	14.6%
Body_Type	object	SUV, Sedan, Hatchback, etc.	0	0.0%
Drivetrain	object	Drive layout (e.g., FWD, AWD)	0	0.0%
Fuel_Efficiency	float64	Fuel economy rating	794	14.4%
Location	object	Location code (e.g., state)	782	14.2%
Selling_Price	int64	Target: selling price	0	0.0%

A sample of the records after feature engineering is shown in Fig 4.2.



Feature engineering completed.

	Make	Model	Year	Fuel_Type	Transmission	Engine_Size	Mileage	Horsepower	Torque	Owners	Accident_History	Service_History	Color	Body_Type	Drivetrain	Fuel_Efficiency	Location	Selling_Price	Car_Age	Mileage_per_Engine
0	Mercedes-Benz	GLE	2024	Petrol	Automatic	2.3	100	186.0	196.0	1	0.0	Full Service	Silver	SUV	AWD	30.0	IL	64140	2	43.478261
1	Hyundai	Tucson	2008	Petrol	Automatic	2.3	387035	189.0	190.0	4	0.0	Full Service	Blue	SUV	FWD	35.0	FL	500	18	168276.086957
2	Volkswagen	Golf	2021	Hybrid	Automatic	1.9	46054	158.0	153.0	1	0.0	Partial Service	Gray	Hatchback	AWD	43.0	NY	16429	5	24238.947368
3	Chevrolet	Tahoe	2005	Petrol	Automatic	2.2	141302	169.0	165.0	5	1.0	No Service	Blue	SUV	AWD	32.0	FL	2199	21	64228.181818
4	Toyota	Camry	2022	Petrol	Automatic	1.9	32813	149.0	141.0	1	0.0	Full Service	Brown	Sedan	FWD	38.0	CA	21792	4	17270.000000

Fig 4.2 Sample records of the dataset after feature engineering

4.3 Pre-processing and Feature Engineering

Cleaning

The duplicate check found no repeated rows. Numerical attributes with missing values (Engine_Size, Horsepower, Torque, Accident_History and Fuel_Efficiency) were filled with the column median, which is robust to the skew present in the data. Categorical attributes (Transmission, Service_History, Color and Location) were filled with the most frequent category. After cleaning, no missing values remained and the shape stayed 5,500 × 18. Image-specific steps such as resizing and flipping, and data augmentation, are not applicable to tabular data.

Feature engineering

Three features were derived from existing attributes (Table 4.3). Price_per_HP was intentionally not created because it divides the selling price by horsepower; using the target to build an input feature would leak the answer into the model and inflate the results.

Table 4.3 Engineered features

Feature	Definition	Purpose
Car_Age	Current year (2026) minus Year	Expresses vehicle age directly
Mileage_per_Engine	Mileage ÷ Engine_Size (zero engine size treated as missing)	Relates usage to engine capacity
First_Owner	1 if Owners = 1, otherwise 0	Flags single-owner vehicles
Price_per_HP (excluded)	Selling_Price ÷ Horsepower	Removed to prevent target leakage

Preprocessing pipeline

After the split, a ColumnTransformer prepares the inputs (Fig 4.3). The 11 numerical features pass through a median imputer. The 9 categorical features pass through a most-frequent imputer and a one-hot encoder that ignores unseen categories. Feature scaling is not applied because tree-based models are insensitive to feature scale.

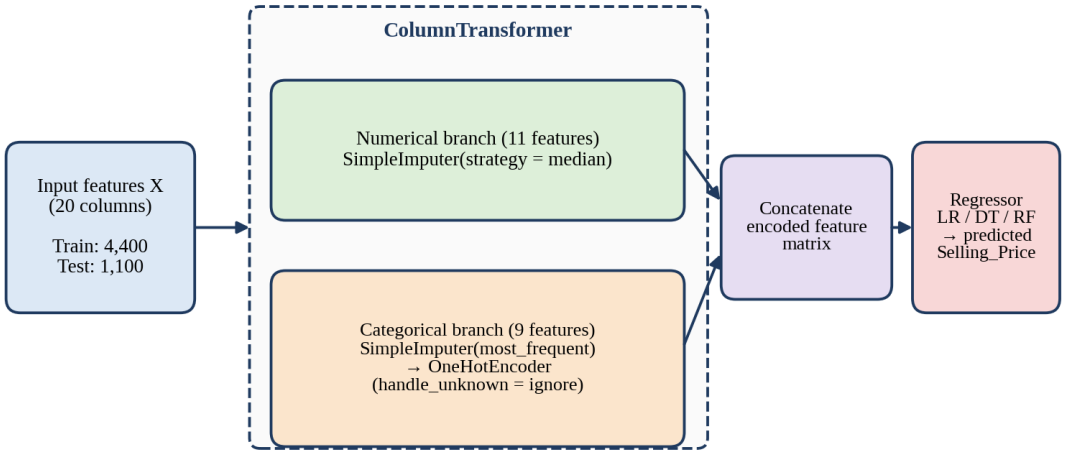


Fig 4.3 Preprocessing and modelling pipeline

4.4 Model / Algorithm

Three regressors were trained on the same prepared features. **Linear Regression** fits a weighted sum of the inputs by ordinary least squares and serves as the baseline. **Decision Tree Regression** recursively splits the feature space to minimise the squared error within each region; its depth was limited to 10 to control overfitting. **Random Forest Regression** [1] trains many trees on bootstrap samples with random feature subsets and averages their outputs, which reduces variance [14]. For T trees $h_1 \dots h_T$ the prediction is given in Eq. (1), and the quality of every model is measured by Eqs. (2)–(4).

$$\hat{y}(x) = (1/T) \cdot \Sigma\, h_t(x), \quad t = 1 \dots T \qquad (1)$$

$$MAE = (1/n)\, \Sigma\, |y_i - \hat{y}_i| \qquad (2)$$

$$RMSE = \sqrt{[(1/n)\, \Sigma\, (y_i - \hat{y}_i)^2]} \qquad (3)$$

$$R^2 = 1 - \Sigma\, (y_i - \hat{y}_i)^2 / \Sigma\, (y_i - \bar{y})^2 \qquad (4)$$

The loss minimised by the tree-based models is the mean squared error; no gradient-based optimiser is involved. Table 4.4 lists the configuration of each model, and Fig 4.4 illustrates the Random Forest.

Table 4.4 Model hyperparameters

Model	Configuration	Loss / fitting
Linear Regression	Default estimator configuration	Least squares
Decision Tree Regression	max_depth = 10, random_state = 42	Squared error
Random Forest Regression	n_estimators = 200, random_state = 42, n_jobs = −1	Squared error
Data split	test_size = 0.20, random_state = 42	4,400 / 1,100

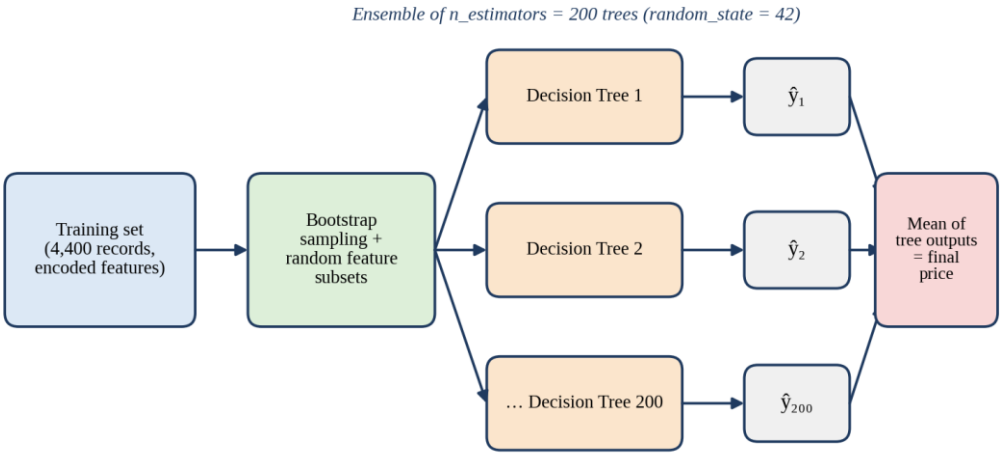


Fig 4.4 Architecture of the Random Forest Regression model

4.5 Tools and Technologies

The system was implemented in Python 3 on Google Colab. pandas and NumPy handle data, matplotlib, seaborn and Plotly produce the charts, and scikit-learn [15] provides the imputers, encoder, pipeline, models, feature-selection functions and metrics. OpenCV, TensorFlow and PyTorch were not needed because the problem involves structured data and classical machine learning models.

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Experimental Setup

All experiments were run in a Google Colab notebook. Table 5.1 summarises the hardware and software environment.

Table 5.1 Hardware and software specifications

Component	Specification
Platform	Google Colab (cloud-hosted notebook)
Processing	Colab CPU runtime; GPU not required (n_jobs = −1 for parallel tree training)
Language	Python 3
Libraries	pandas, NumPy, matplotlib, seaborn, Plotly, scikit-learn
Input / output files	automobile_dataset.csv; cleaned_automobile_dataset.csv; automobile_price_predictions.csv; automobile_model_metrics.csv

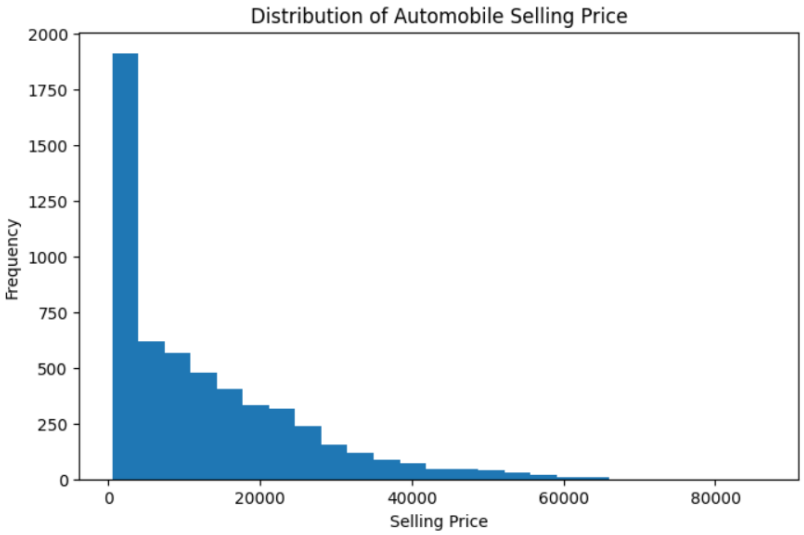
5.2 Exploratory Data Analysis

Fig 5.1(a) shows the output of the inspection step: nine attributes have between 782 (Location) and 855 (Service_History) missing entries, about 14–15.5% of the rows each, while the remaining nine attributes are complete. Fig 5.1(b) shows that Selling_Price is strongly right-skewed. Most vehicles sell below about 20,000, a long tail extends beyond 60,000, and a single record lies near 86,000. Expensive vehicles are therefore rare, which makes them harder to predict.

Missing values:

	0
Service_History	855
Engine_Size	844
Transmission	827
Accident_History	812
Color	803
Horsepower	800
Torque	799
Fuel_Efficiency	794
Location	782
Make	0
Model	0
Mileage	0
Fuel_Type	0
Year	0
Owners	0
Body_Type	0
Drivetrain	0
Selling_Price	0

(a) Missing values

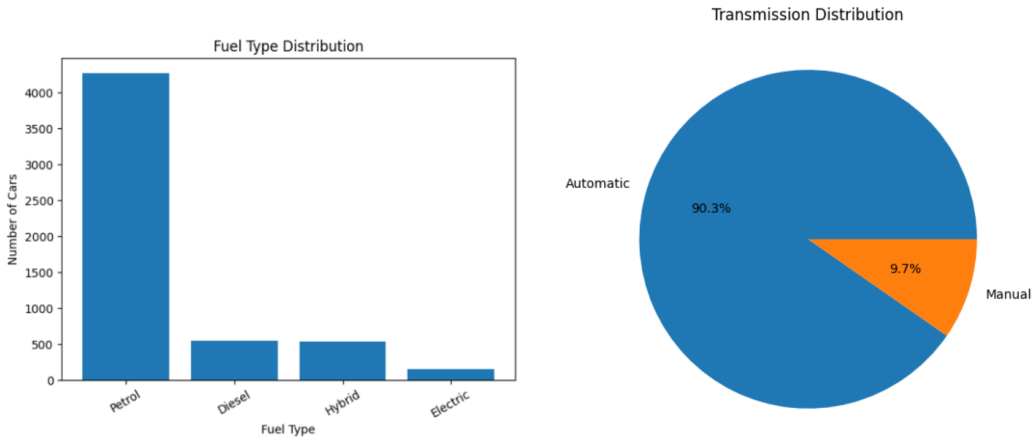


(b) Selling price distribution

Fig 5.1 Missing values per attribute and distribution of Selling_Price

Petrol vehicles dominate the data (roughly 4,300 of 5,500 records), whereas diesel and hybrid vehicles account for about 500–550 each and electric vehicles for about 150 (Fig 5.2a). Automatic transmission accounts for 90.3% of the records and manual for 9.7% (Fig

5.2b); because 827 missing transmission values were filled with the most frequent category, the automatic share is partly a product of imputation.



(a) Fuel type *(b) Transmission*

Fig 5.2 Distribution of fuel type and transmission

The relationship between usage and price is clear in Fig 5.3: prices fall as mileage rises, vehicles above 40,000 occur only at low mileage, and cars beyond 300,000 sell at the bottom of the price range. The interactive Plotly version (Fig 5.4) colours points by fuel type and exposes make and model on hover, which allows individual records to be inspected.

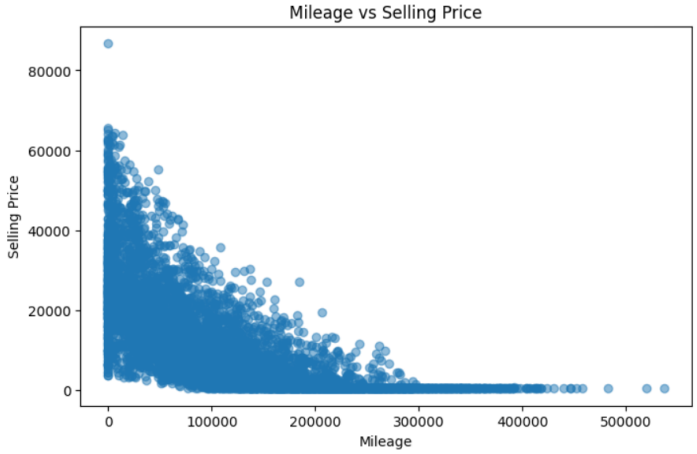


Fig 5.3 Mileage versus selling price

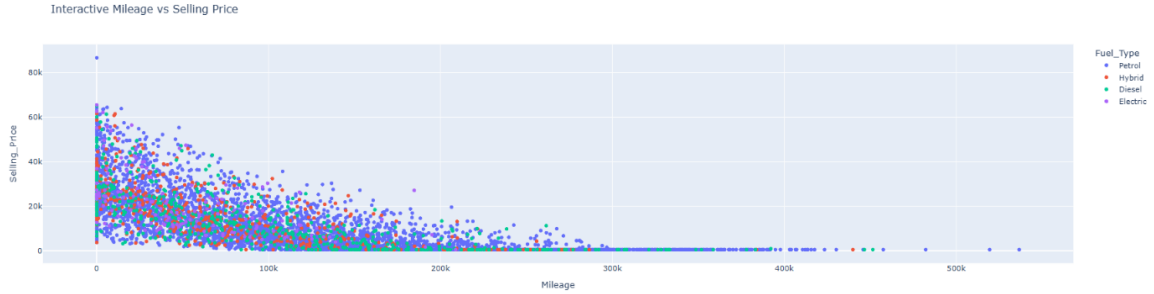


Fig 5.4 Interactive mileage versus selling price plot (colour = fuel type)

The correlation heatmap in Fig 5.5 quantifies the associations. Selling_Price correlates positively with Year (0.80) and negatively with Mileage (−0.68), Owners (−0.67) and Accident_History (−0.35). Year is also strongly related to Owners (−0.83) and Mileage

(−0.75), so newer cars tend to have fewer owners and lower mileage. Horsepower and Torque are strongly related to each other (0.83), but Engine_Size, Horsepower, Torque and Fuel_Efficiency have almost no linear correlation with price ($|r| \leq 0.12$). These are statistical associations, not proof of causation.

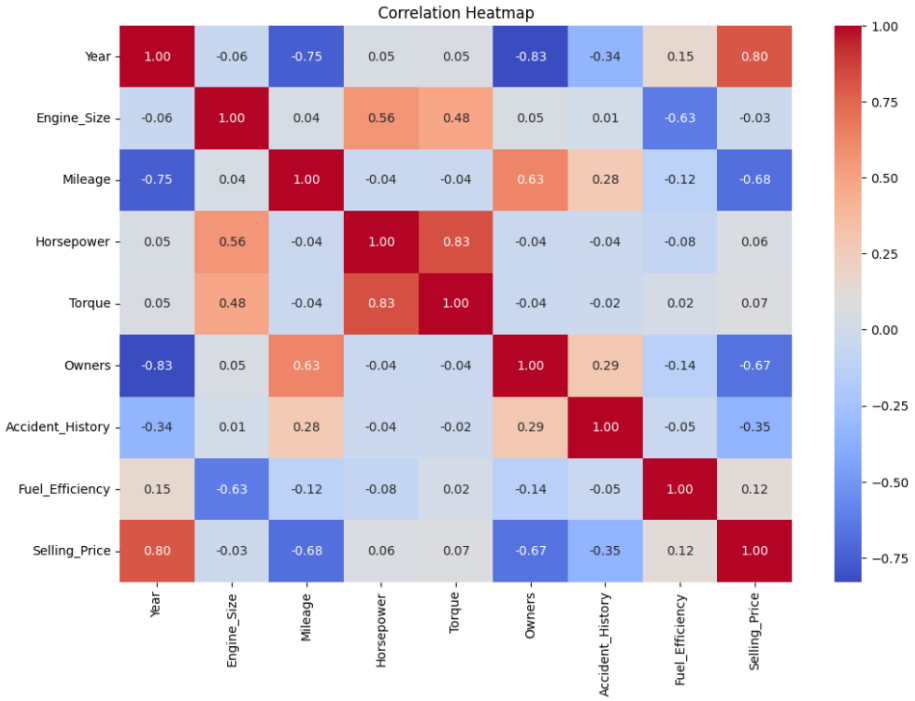


Fig 5.5 Correlation heatmap of the numerical attributes

5.3 Feature Selection Results

Three regression-based methods ranked the 20 features (Fig 5.6). Categorical columns were integer-coded only for this analysis, so their correlation and F-scores should be read with caution; the ranking was used for insight and all 20 features were retained for modelling. Car_Age and Year obtain identical correlation (0.8049) and F-score (10,117.63) because Car_Age is an exact linear function of Year. Mileage, First_Owner, Owners and Mileage_per_Engine are consistently ranked high, showing that the engineered features carry useful information. Table 5.2 compares the leading features across the methods.

	Absolute Correlation
Car_Age	0.804933
Year	0.804933
Mileage	0.682883
First_Owner	0.682265
Owners	0.669046
Mileage_per_Engine	0.630522
Accident_History	0.347294
Transmission	0.243412
Fuel_Type	0.132468
Fuel_Efficiency	0.118963

(a) Absolute correlation

	Feature	F Score	p-value
17	Car_Age	10117.629328	0.000000e+00
2	Year	10117.629328	0.000000e+00
6	Mileage	4804.220023	0.000000e+00
19	First_Owner	4787.966858	0.000000e+00
9	Owners	4455.343134	0.000000e+00
18	Mileage_per_Engine	3628.184590	0.000000e+00
10	Accident_History	754.083702	1.099599e-155
4	Transmission	346.270248	5.314725e-75
3	Fuel_Type	98.200238	5.884102e-23
15	Fuel_Efficiency	78.925259	8.600369e-19

(b) F-score

	Feature	MI Score
2	Year	0.723398
17	Car_Age	0.719985
6	Mileage	0.587521
18	Mileage_per_Engine	0.527912
9	Owners	0.398481
19	First_Owner	0.302089
1	Model	0.115194
0	Make	0.089351
10	Accident_History	0.067075
4	Transmission	0.054742

(c) Mutual information

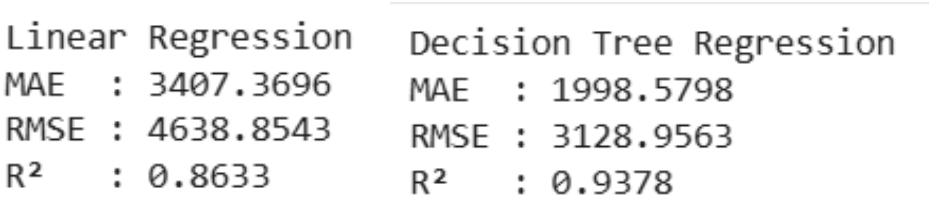
Fig 5.6 Top ranked features by correlation, F-score and mutual information

Table 5.2 Top five features under each selection method

Rank	Absolute correlation	F-score	Mutual information	RF importance (Sec. 5.6)
1	Car_Age (0.8049)	Car_Age (10,117.63)	Year (0.7234)	Car_Age (0.3487)
2	Year (0.8049)	Year (10,117.63)	Car_Age (0.7200)	Year (0.3466)
3	Mileage (0.6829)	Mileage (4,804.22)	Mileage (0.5875)	Make_Mercedes-Benz (0.0584)
4	First_Owner (0.6823)	First_Owner (4,787.97)	Mileage_per_Engine (0.5279)	Make_BMW (0.0575)
5	Owners (0.6690)	Owners (4,455.34)	Owners (0.3985)	Make_Audi (0.0556)

5.4 Model Performance

The three models were trained on the 4,400 training records and evaluated on the 1,100 test records. Fig 5.7 shows the console output for Linear Regression and Decision Tree Regression, and Table 5.3 gives the final comparison, which is also shown as the notebook output in Fig 5.8. In place of the confusion matrix, accuracy curves and mAP used in classification and detection projects, the regression metrics MAE, RMSE and R^2 are reported.



(a) Linear Regression *(b) Decision Tree Regression*

Fig 5.7 Evaluation output of the baseline models

Table 5.3 Performance comparison of the regression models

Model	MAE	RMSE	R^2
Random Forest Regression	1349.7979	2113.4888	0.9716
Decision Tree Regression	1998.5798	3128.9563	0.9378
Linear Regression	3407.3696	4638.8543	0.8633

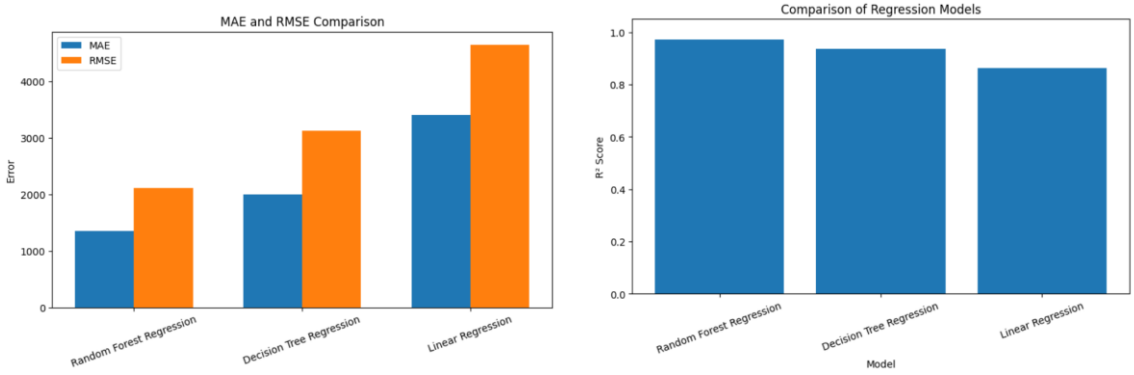
	Model	MAE	RMSE	R2
0	Random Forest Regression	1349.797891	2113.488803	0.971626
1	Decision Tree Regression	1998.579761	3128.956280	0.937811
2	Linear Regression	3407.369630	4638.854272	0.863310

Fig 5.8 Notebook output of the final model comparison

Random Forest has the lowest errors and the highest R^2. Relative to Linear Regression it lowers MAE by 60.4% and RMSE by 54.4%; relative to the depth-limited Decision Tree it lowers both by about 32.5% (Table 5.4). The gain over the linear baseline confirms that the price–attribute relationship is non-linear, and the gain over a single tree shows the benefit of averaging 200 trees.

Table 5.4 Improvement of Random Forest over the baseline models

Compared with	MAE reduction	RMSE reduction	R^2 gain
Linear Regression	60.4%	54.4%	+0.1083
Decision Tree Regression	32.5%	32.5%	+0.0338



(a) MAE and RMSE *(b) R^2 score*

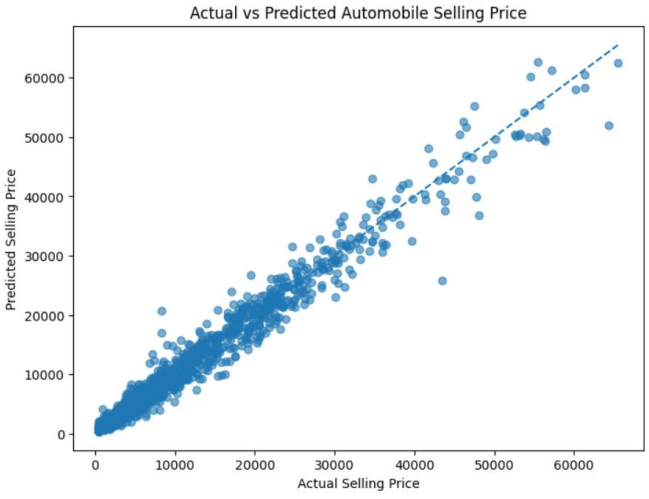
Fig 5.9 Graphical comparison of the three regression models

5.5 Prediction Analysis

Fig 5.10(a) lists ten test predictions. Several are very close (500 predicted as 500.00; 18,875 as 19,764.55), while the largest absolute error among them is the 31,208 vehicle, under-predicted by about 6,384. Relative errors are largest for cheap vehicles, where a small absolute difference is a large percentage. In the actual-versus-predicted plot (Fig 5.10b) the points follow the diagonal closely, with more scatter and a tendency to fall below the line for prices above about 40,000.

	Actual_Selling_Price	Predicted_Selling_Price
0	3837	4162.360
1	500	500.000
2	10127	12049.585
3	5357	7604.300
4	18121	19539.100
5	18875	19764.545
6	8926	7864.575
7	857	1079.420
8	5886	6450.820
9	31208	24823.795

(a) Sample predictions



(b) Actual vs predicted

Fig 5.10 Sample predictions and actual versus predicted selling price

The residual plot (Fig 5.11) is centred on zero without an obvious curve, which suggests the model is not systematically biased. However, the spread of the residuals widens as the predicted price increases (roughly ±5,000 at mid prices, and individual errors of about

+17,500 and −12,500), so the model is less reliable for expensive vehicles. This matches the scarcity of high-priced records seen in Fig 5.1(b).

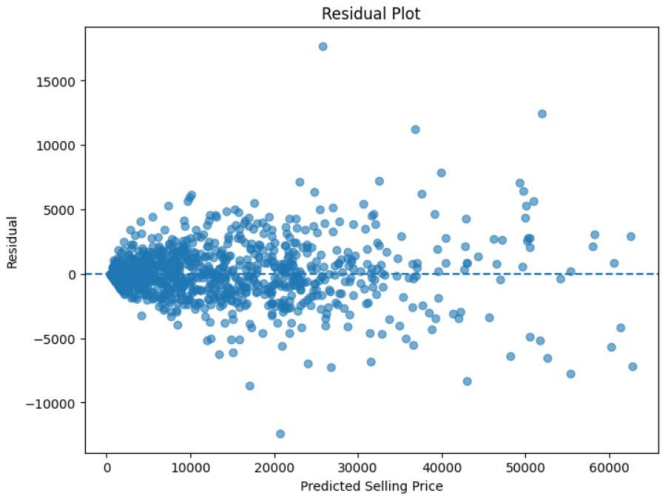


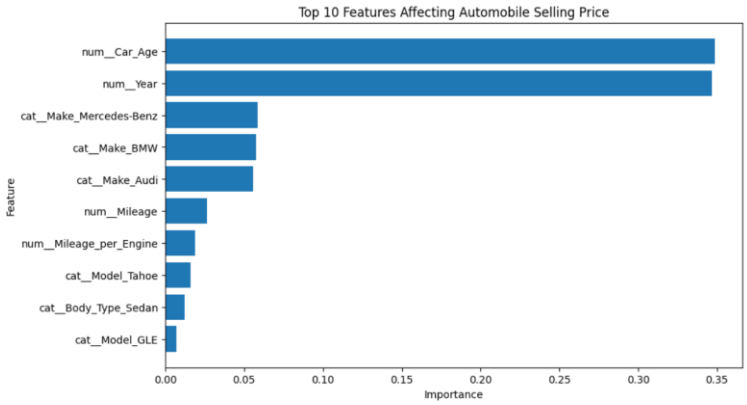
Fig 5.11 Residual plot of the Random Forest model

5.6 Feature Importance

Fig 5.12 shows which inputs the fitted Random Forest relies on most. Car_Age (0.3487) and Year (0.3466) together account for about 69.5% of the total importance; because they carry the same information, the forest shares the credit between them. Premium manufacturers follow, with Mercedes-Benz, BMW and Audi each contributing about 0.056–0.058. Mileage (0.0265) and Mileage_per_Engine (0.0191) come next, followed by specific models and body types such as Tahoe, Sedan and GLE. Horsepower, Fuel_Efficiency and Torque each contribute less than 0.003. These values describe how the fitted model uses its encoded features; they do not prove that a feature independently causes a price change.

	Feature	Importance
8	num__Car_Age	0.348681
0	num__Year	0.346601
17	cat__Make_Mercedes-Benz	0.058398
12	cat__Make_BMW	0.057547
11	cat__Make_Audi	0.055607
2	num__Mileage	0.026466
9	num__Mileage_per_Engine	0.019086
56	cat__Model_Tahoe	0.016137
81	cat__Body_Type_Sedan	0.012236
40	cat__Model_GLE	0.007298
23	cat__Model_A4	0.003444
60	cat__Model_X5	0.003413
3	num__Horsepower	0.002839
7	num__Fuel_Efficiency	0.002551
4	num__Torque	0.002484

(a) Importance values



(b) Top 10 features

Fig 5.12 Feature importance of the Random Forest model

5.7 Discussion and Comparison

Random Forest was the best of the three models on every metric, which is consistent with earlier findings that Random Forest is effective for resale-price forecasting [2] and that good feature representation matters [13]. The results are not compared numerically with the published studies because their datasets, features and settings differ. The predictions and metrics were saved as CSV files for reproducibility (Fig 5.13).

```
Saved: automobile_price_predictions.csv
Saved: automobile_model_metrics.csv
```

	Metric	Value
0	MAE	1349.797891
1	RMSE	2113.488803
2	R2	0.971626

Fig 5.13 Saved prediction and metric files

Several factors should be kept in mind when interpreting the results. First, imputation was applied to the full dataset before the train–test split, so some information from the test records may have influenced preprocessing. Second, the results come from a single split without cross-validation or an external dataset. Third, Car_Age is computed from the current year, so it changes with the run date, and it duplicates Year. Fourth, the Decision Tree and Random Forest were not tuned beyond the stated settings. The reported performance is therefore best treated as an estimate for this dataset rather than for entirely unseen data.

CHAPTER 6
CONCLUSION

6.1 Summary of Results

This project built an end-to-end workflow for predicting used-automobile selling prices from 5,500 records. The data were cleaned, explored, enriched with three engineered features, ranked with three selection methods, and used to train Linear Regression, Decision Tree Regression and Random Forest Regression. Random Forest achieved the best results on the held-out test set, with an MAE of 1349.7979, an RMSE of 2113.4888 and an R^2 of 0.9716, compared with R^2 values of 0.9378 for the Decision Tree and 0.8633 for Linear Regression. Vehicle age and manufacture year were the leading predictors, followed by premium manufacturers and mileage. The results support the use of ensemble tree models for automobile price estimation on this dataset.

6.2 Limitations

- Imputation was performed on the whole dataset before splitting, which may let test information influence preprocessing.
- Only one train–test split was used; no cross-validation or external validation was carried out.
- Year and Car_Age are redundant, so their importance values must be interpreted together.
- Errors grow for high-priced vehicles, where training examples are scarce.
- Models were not hyperparameter-tuned, and no deployment or timing measurements were made.

6.3 Future Work

Future work can fit all data-dependent preprocessing inside the training folds and evaluate the models with k-fold cross-validation. Hyperparameter tuning and additional ensemble methods such as gradient boosting may improve accuracy further. Testing on an independent dataset from another source or period would show how well the model generalises, and a log-transformed target or separate treatment of premium vehicles may reduce the errors at high prices. Finally, the trained pipeline can be deployed as a simple web application so that buyers and sellers can obtain instant price estimates.

CHAPTER 7
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